

Jayesh Mahajan

AI & ML Engineer – Proficient in Machine Learning, NLP, Deep Learning, and Generative AI (LangChain, RAG, LLMs)

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Professional Summary

AI/ML Engineer with hands-on experience building **LLM-powered applications, RAG systems, and multimodal AI pipelines**. Skilled in developing scalable AI solutions using **Python, FastAPI, LangChain, Azure OpenAI, and vector databases**. Experienced in end-to-end machine learning workflows including **data preprocessing, model development, and deployment of production-ready AI systems**.

Education

PES Modern College of Engineering

Bachelor of Engineering (B.E.) in Artificial Intelligence & Machine Learning | CGPA: 8.5

Pune, MH

2022 – 2026

Technical Skills

- **Programming:** Python, SQL, Java, JavaScript
- **Generative AI & LLMs:** Azure OpenAI, OpenAI API, Llama 3, Mistral, Hugging Face Transformers, LangChain, LlamaIndex, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings
- **AI & Machine Learning:** Supervised Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Exploratory Data Analysis (EDA), Feature Engineering, Model Evaluation
- **AI Systems:** OCR (Tesseract, EasyOCR), Whisper STT, Document Processing (PyPDF2, pdf2image), Multimodal AI Pipelines
- **Backend & APIs:** FastAPI, Flask, REST APIs, Authentication, RBAC Systems
- **Databases & Cloud:** PostgreSQL, Azure Cosmos DB, Azure Blob Storage, Azure AI Search, Supabase, FAISS, Chroma
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, NLTK, spaCy, OpenCV
- **Applications:** Streamlit, React (Basic), HTML, JavaScript, Expo
- **Tools:** Git, GitHub, CI/CD Workflows, Docker
- **Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Statistical Modeling, Neural Networks

Experience

AI Intern

Aditya Birla Group | UltraTech Cement Limited

Jan 2026 – Apr 2026

Mumbai, India

Project: SkillSpotter – AI HR Analytics & CV Screening Platform

- Developed an AI-powered hiring platform used by **UltraTech HR teams across multiple locations** to match resumes with job descriptions and generate candidate fit scores.
- Integrated **Azure Cosmos DB, Azure Blob Storage, and Azure AI Search** to manage **10000+ CV records** and enable semantic candidate retrieval.
- Optimized the LLM pipeline by migrating from **GPT-4o to GPT-5**, reducing response latency by **30%+** for recruiter screening workflows.
- Generated automated **Excel and PDF hiring reports**, enabling faster recruiter evaluation and standardized candidate scoring.

Project: RCO Management System – Renewable Energy Compliance Platform

- Developed an enterprise platform to automate **Renewable Consumption Obligation (RCO)** tracking across **72+ UltraTech plants**, replacing manual Excel workflows.
- Built a **full-stack application using React, FastAPI, and PostgreSQL** with REST APIs and role-based access control.
- Implemented automated compliance calculations for fossil energy usage, renewable energy targets, and REC conversion.

AI Intern

FlatZ – AI Community Platform

Sep 2025 – Nov 2025

Remote

- Developed an **AI-powered community assistant** used by **1000+ residents across 3 residential communities** for visitor management and maintenance workflows.
- Built backend services using **FastAPI and PostgreSQL** with **LLM-based intent classification** and action routing for community operations.
- Implemented conversational features including visitor approvals, **QR-based entry**, and natural-language visitor registration.
- Designed an **AI-driven maintenance assistant** enabling users to query request history, filter requests, and retrieve summaries using natural language.
- Collaborated via **Git pull requests** with automated DevOps CI checks, contributing to production-ready mobile builds tested using **Expo Go**.

- Implemented an end-to-end **machine learning pipeline** using MIMIC-IV healthcare data to predict patient mortality and hospital length of stay.
- Performed extensive **data preprocessing** including cleaning, missing value imputation, outlier detection, and normalization.
- Conducted **Exploratory Data Analysis (EDA)** to identify feature correlations and trends.
- Developed **Logistic Regression, Random Forest, and XGBoost models**, improving predictive accuracy by **7%**.
- Built **modular ML pipelines** for scalable experimentation and reproducible model development.

Projects

MIRA – Multimodal Intelligent Reasoning Agent

FastAPI, Python, LangChain, HuggingFace, Ollama, Whisper, OCR

- Designed an **agentic AI system** capable of reasoning across **text, images, PDFs, and audio** inputs.
- Implemented multimodal ingestion pipelines using **Tesseract OCR, PyPDF2/pdf2image, and Whisper STT**.
- Built an **intent detection and task orchestration layer** enabling autonomous extraction, summarization, and query answering.
- Integrated **LLaMA 3.2 via Ollama and HuggingFace Transformers** with **LangChain** for reasoning, memory, and tool execution.
- Exposed the agent through a **FastAPI backend** with an interactive interface enabling real-time multimodal interactions.

Juri AI – RAG-Based Chatbot (LegalTech Domain)

Python, LangChain, ChromaDB, Groq LLaMA 3.1, Streamlit

- Developed a legal assistance chatbot using **RAG architecture** with **LLaMA 3.1 and ChromaDB**.
- Processed legal documents into structured chunks enabling **semantic retrieval and query interpretation**.
- Implemented a document retrieval and response generation pipeline using **LangChain**.
- Built an interactive **Streamlit interface** allowing conversational legal queries.
- Optimized prompt design and retrieval parameters to improve response accuracy.
- Implemented vector similarity search with **ChromaDB embeddings** to retrieve legally relevant contextual passages.

StoryTails – AI-Powered Text-to-Video Story Generator *Python, Llama 3, SDXL, Stable Video Diffusion, Coqui XTTS, FFmpeg*

- Developed an end-to-end **AI storytelling pipeline** converting text prompts into animated videos.
- Implemented LLM-driven scene extraction using **Llama 3**.
- Generated visuals using **SDXL and Stable Video Diffusion**.
- Integrated neural TTS (**Coqui XTTS**) and synchronized narration using **FFmpeg**.
- Built a **Streamlit interface** for story generation and video preview/download.

Bank Marketing Campaign Classification – Predictive Analytics Model

Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Developed a **predictive model** to classify customer responses for bank marketing campaigns.
- Performed **EDA, feature engineering, encoding, and feature selection**.
- Trained supervised models including **Logistic Regression, Decision Tree, and Random Forest**.
- Implemented a reusable **ML pipeline** for automated preprocessing and evaluation.
- Generated visualization dashboards highlighting campaign conversion trends.

Certifications

- Oracle Cloud Infrastructure Certified Data Science Professional – **Oracle**
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional – **Oracle**
- Python for Data Science – **IBM**
- Data Visualization – **TATA Forage**
- Software Engineering – **TATA Forage**

Soft Skills

- Strong **leadership** and **team collaboration** skills with the ability to adapt quickly in dynamic project environments.
- Excellent **communication, presentation, and problem-solving** abilities with a logical and analytical mindset.